

5.4.3. Discovery by Commissioner

How a Commissioner discovers a commissionable device depends on the communication interfaces that the device and the Commissioner support (see [Section 5.4.2.1, “Announcement and Discovery Using Multiple Technologies”](#)). Though not all communication interfaces must be supported by every device (see [Table 58, “Discovery Capabilities Bitmask”](#)), a Commissioner SHALL support Commissioning (see [Section 5.5, “Commissioning Flows”](#)) using existing IP network and over BLE (if equipped with a BLE interface). Furthermore, commissioners SHOULD support commissioning over Wi-Fi Public Action Frame (if equipped with a Wi-Fi interface), and SHOULD support NFC-based Commissioning (if equipped with an NFC interface).

A Commissioner that is aware of the device’s Discovery Capability Bitmask SHALL initiate Device Discovery on all of the communication interfaces that are supported by both the Commissioner and the device. A Commissioner that is unaware of the device’s Discovery Capability Bitmask SHALL initiate Device Discovery on all of the networking technologies it supports out of BLE, Wi-Fi Public Action Frame, and DNS-SD network discovery on IP networks.

Commissioners SHALL always support discovering a device using DNS-based Service Discovery (DNS-SD) for commissioning, irrespective of the [Discovery Capabilities Bitmask](#) specified in the [Onboarding Payload Contents](#).

The following sections detail Commissioner behavior for each of these networking technologies. Though a QR code, NFC Tag, or Manual Pairing code may be scanned, tapped or entered prior to discovery, it is not required to do so. However, after scan/tap/entry of the code, the Discriminator, VID and PID elements are available to ensure that the intended device is discovered before proceeding to the connection phase of commissioning.

5.4.3.1. Using BLE

Commissioners SHALL implement the role of a GAP Central. To discover a commissionable device advertising over BLE, a Commissioner SHALL perform a BLE scan across all three advertising channels with a sufficient dwell time, interval, and overall duration of scan. In order to promote quick discovery it is recommended that a Commissioner scan as aggressively as possible within the Commissioner device functionality/UX constraints. In addition, if manufacturer-specific data is not needed, a passive scan (*i.e.*, one that only listens for Advertisement PDUs and does not issue Scan Request PDUs) SHOULD be used.

If discovery procedure is user initiated the scan interval SHOULD be set between 30 ms and 60 ms, and the scan window SHOULD be set to 30 ms. If discovery procedure is not user initiated (*i.e.*, the Commissioner is scanning in the background), the device may use more relaxed scan, for example, the scan interval set to 1.28 seconds and scan window set to 11.25 ms.

NOTE

Recommended values are defined in Appendix A: Timers and Constants of [Bluetooth® Core Specification 4.2 \(amended\)](#), Vol 3, Part C.

5.4.3.2. Using Wi-Fi Public Action Frame

To discover a commissionable device advertising using Wi-Fi Public Action Frame (see [Section 5.4.2.6, “Using Wi-Fi Public Action Frame”](#)), a Commissioner that supports commissioning using Wi-